

Sample Exam

Medical Image Analysis

Student name:_____

Immatriculation number:_____

Exam Questions

Note: All the questions have equal weight.

1. In axial 2D acquisitions the slices lie in the transverse plane and are stacked along the head-to-toe direction. Voxel sizes are given as in-plane (two values) and through-plane (the slice thickness), and the field-of-view is image size $\times$ voxel size.

An axial acquisition uses in-plane voxel size $0.7 \times 0.7 \text{ mm}^2$ with 320 pixels in both directions, and through-plane voxel size 4.0 mm with 44 slices.

- (a) What is the field-of-view in the head-to-toe, left-to-right and anterior-to-posterior directions, respectively?
- (b) The anatomy that has to be covered extends 210 mm in the head-to-toe direction. Does this protocol cover it? If not, how many slices are needed at the same slice thickness, and by what factor does the acquisition time change if it is proportional to the number of slices?
- (c) Compute the anisotropy ratio of the voxels. This volume is then fed, without resampling, to a 3D convolutional network that was trained on 1.0 mm isotropic data. What goes wrong?

2. In atlas-based segmentation the atlas is registered to the image and then used inside the mixture model. Which of the two posteriors below is the one used by atlas-based segmentation, as opposed to plain EM with a Gaussian mixture model? π_n denotes a mixture weight estimated from the image itself. Please circle the correct choice.

$$\begin{aligned} p(c(x)=n \mid I(x)) &\propto \mathcal{N}(I(x) \mid \mu_n, \Sigma_n) \pi_n \\ p(c(x)=n \mid I(x)) &\propto \mathcal{N}(I(x) \mid \mu_n, \Sigma_n) p_{atlas}(c(x)=n) \end{aligned}$$

3. Registration problems are classified along two axes: whether the two images come from the *same or different subjects*, and whether they were acquired with the *same or different modalities*.

(a) Which of the four combinations is the easiest and which is the hardest, and why?

(b) Each axis primarily determines one component of the registration algorithm. Which axis determines which, and why?

4. Please indicate whether the statement is true or false by circling the correct choice in each of the following.

(a) True False The 3×3 mean filter is separable, $S_3 = \frac{1}{9}[1, 1, 1] * [1, 1, 1]^T$, which makes it cheaper to apply than a general 3×3 kernel.

(b) True False A Gaussian has finite support, so a 3×3 structural element represents a Gaussian filter exactly.

(c) True False Binomial filters are built by iterated convolution of $[1, 1]$ and have better properties in the frequency domain than mean filtering.

(d) True False Increasing σ of a Gaussian filter smooths more, but the size of structural element required is independent of σ .

5. A classification network takes an input image $x \in \mathbb{R}^{64 \times 64}$ with a single channel and maps it through hidden layers of 256, 128 and 64 units to 16 output classes. Consider only the *first* link, from the input to the 256-unit layer, and ignore biases throughout.

- (a) How many weights does this link have if it is fully connected?
- (b) How many weights does it have if it is instead a valid 5×5 convolution producing 256 output channels from the single input channel?
- (c) What is the ratio, and what property of the convolutional layer explains it?
6. In atlas-based segmentation the model is $p(I(x) \mid \theta) = \sum_n \mathcal{N}(I(x) \mid \mu_n, \Sigma_n) p_{atlas}(c(x)=n)$ with $\theta = \{\mu_n, \Sigma_n\}_{n=1}^N$. Please explain why the intensity parameters μ_n and Σ_n still have to be estimated for every new MR image, but could in principle be fixed in advance for CT. Then state what else the atlas gives you compared with plain EM. You do not need to write down the equations.
7. Which of the following statements about interpolation is **inaccurate**?
- ☐ (A) Bilinear interpolation can equivalently be written as the parametric surface $I(x_1, x_2) = a + bx_1 + cx_2 + dx_1x_2$, whose four unknowns are fixed by the intensities at the four surrounding pixels.
 - ☐ (B) The trilinear extension adds the remaining monomials to give eight unknowns, fixed by the eight corner voxels of the cell containing the query point.

- (C) Bicubic interpolation has 16 unknowns, determined by the four corner intensities together with their first and mixed derivatives — and since those derivatives must themselves be approximated from further pixels, bicubic effectively uses more than four samples.
 - (D) Because bilinear interpolation is a weighted average of the surrounding values, applying it repeatedly during an iterative registration leaves the effective spatial resolution unchanged.
8. Which of the following statements about the loss functions used for image restoration is **incorrect**?
- (A) MAE, MSE, NMSE, PSNR and SSIM are all computed directly between the prediction and the ground-truth image.
 - (B) SSIM compares many local patches of the two images and combines three components — luminance, contrast and structure.
 - (C) A perceptual loss passes both images through a previously trained network and compares intermediate features; that feature network must have been trained on the same task and the same modality as the restoration model, otherwise the comparison is meaningless.
 - (D) An adversarial loss is a distributional distance defined over *sets* of samples rather than between two paired images.
9. A semi-supervised auxiliary loss enforces consistency under spatial transformations. Two transformations ϕ_1, ϕ_2 are drawn per image and the loss is $\mathcal{L}(\phi_2 \circ \phi_1^{-1} \circ f(\phi_1 \circ x; \theta), f(\phi_2 \circ x; \theta))$. Which of the following statements is **incorrect**?
- (A) The loss requires no labels, so it can be evaluated on unlabelled images.
 - (B) It encodes the principle that if $x \Leftrightarrow y$ then $\phi \circ x \Leftrightarrow \phi \circ y$, which is the same principle that underlies data augmentation.
 - (C) The inverse ϕ_1^{-1} appears so that the two predictions are brought into a common spatial frame; without it they would sit in different frames and could not be compared pixel by pixel.
 - (D) Because the loss is defined on unlabelled data, it replaces the supervised loss rather than being added to it.
10. When the network parameters are sampled from their posterior rather than optimised to a point estimate, the sampled parameters can perform noticeably worse on the task than the MAP estimate. What is the explanation given in the lecture?
- (A) In high dimensions the posterior is always unimodal, so every sample collapses onto the mean and no diversity is possible.
 - (B) For an isotropic Gaussian in d dimensions the expected distance of a sample from the mean grows like $\sqrt{d}$, so samples concentrate away from the mode — and since the likelihood $p(D \mid \theta, M)$ is the task loss, moving away from the mode means moving away from well-performing parameters.
 - (C) High-dimensional posteriors cannot be sampled at all, so any apparent sample is an artefact of the random number generator.
 - (D) In high dimensions the task loss becomes independent of the parameters, so performance is essentially random.

- ☐ (E) Sampling error decreases with dimension, so the effect disappears entirely for large networks.
11. A Vision Transformer is applied to images of size 224×224 with 3 channels. It uses a patch size of $P = 16$, an embedding dimension of $D = 768$, and 12 attention heads. Recall that the image is split into $N = HW/P^2$ patches, each flattened and mapped by a patch-embedding matrix $E \in \mathbb{R}^{(P^2 C) \times D}$, and that a learnable class token is prepended to the sequence. Circle True or False:
- (a) True False The number of patch tokens is $N = 196$.
 - (b) True False The patch-embedding matrix E is 768×768 and therefore contains 589 824 weights.
 - (c) True False After the class token is added, the sequence entering the encoder still has length 196, because the class token takes the place of one patch.
 - (d) True False The attention matrix QK^T of a single head has $N \times N$ entries, so its size grows linearly with the number of tokens.
12. Which of the following statements on self-supervised learning are correct (multiple answers are possible)?
- ☐ (A) Contrastive learning needs labelled examples in order to decide which pairs of images should be treated as positive and which as negative.
 - ☐ (B) A pre-text task is a task that requires only images; the network trained on it is then fine-tuned with very few labelled examples on the task you actually care about.
 - ☐ (C) In the *local* contrastive loss the transformations must either preserve the patch correspondences or be undone before the corresponding patches are compared.
 - ☐ (D) Foundation models in this setting are obtained by supervised training at very large scale on carefully curated labelled data sets.
 - ☐ (E) The InfoNCE loss maximises the similarity between two views of the same image through its numerator, and minimises the similarity to the samples of the negative set through its denominator.